Group Assignment 2: TrailWatch Wildlife Media

COSC 5606 – Database Management Systems

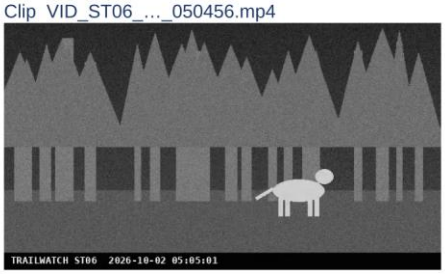
Case study

TrailWatch runs 16 wildlife stations along Lake Superior and the Algoma highlands. Each station has a trail camera, a sound recorder and a small sensor. Rangers log what they see in an Excel workbook. After one month there are **2,330 sighting rows, 151 photos, 13 video clips, 24 audio recordings and about 60,700 sensor readings in JSON**, all sitting in folders. The dataset `TrailWatch_Dataset.zip` is on the LMS; read its `README.md` first. The data is realistic, which means it is messy: species typed several ways, mixed date formats, duplicate rows, and a few broken file links. You must find and handle these yourself.

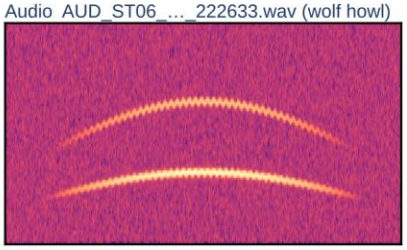
Photo IMG_ST12_..._033214.jpg



Clip VID_ST06_..._050456.mp4



Audio AUD_ST06_..._222633.wav (wolf howl)



The main challenge is the media. A biologist opens a sighting and wants to **see its photo and play its clip in a few seconds**. Where should those files live: inside the database as BLOBs, in a folder with only the path in the database, or in a cloud storage bucket?

Tools. MySQL 8 (MySQL Workbench or the `mysql` command line, whichever you like) and **Supabase**. You do **not** need to install PostgreSQL: Supabase is PostgreSQL run for you in the cloud. A free project at supabase.com is enough for this data (500 MB database, 1 GB file storage, 50 MB per file; the largest file here is 27 MB).

Sample code in `starter/`: `mysql_examples.sql` (load the JSON lines, store a file as a BLOB, time a query) and `supabase_load.py` (load the JSON lines into Supabase and upload every media file to a Storage bucket in one run). Adapt them to your own design.

1. Design the database (2 marks)

Find the best design for this data yourself. Aim for **no more than five tables**. Decide:

(a) which data becomes ordinary tables and columns, and which stays as JSON;

(b) how one sighting connects to its photos, clips and recordings, and where their bytes live;

(c) how to clean species names, station IDs and dates, and remove duplicate sightings.

Draw the ER diagram with primary and foreign keys, and justify each choice in two or three lines. Create your tables in MySQL **and** in Supabase.

2. Store, retrieve and show the media (3 marks)

Load **all 188 media files** three ways, then show that each one works:

A. **MySQL BLOB:** store the bytes in a `LONGBLOB` column, for example with `LOAD_FILE()` (the files must be in the folder named by `secure_file_priv`).

B. **MySQL path:** keep the files in a folder and store only each file's path in the database.

C. **Supabase:** upload the files to a Storage bucket and store each file's URL in the database.

***For each way: time the load, record the database size, and time how long it takes to **retrieve the largest clip** (`VID_ST05_20260910_223205_HD.mp4`). Then **display** the photo and **play** the clip for sighting **TW-00019**, & include screenshot of each. (Hint: Workbench can show a BLOB image through "Open Value in Editor", but it cannot play video. What does that tell you?)*

3. Query and measure (3 marks)

Run each query in MySQL and in Supabase. Run each one **three times** and report the middle time (Workbench: "Duration" in the Output panel; Supabase: `EXPLAIN ANALYZE`, "Execution Time"). Then add an index that helps, run it again, and report both times.

Q1. Sighting **TW-00019**: station name, species and every linked media file.

Q2. Number of **Gray Wolf** sightings at **boreal** stations from 15 September 00:00 to 5 October 23:59, 2026 (all spellings of the species count).

Q3. Number of sensor readings taken while the **battery was below 25%** from 1 to 10 October 2026 (UTC). Ignore motion events and count a re-sent packet once.

Q4. Number of sightings whose notes mention **tracks** or **scat**.

Q5. Fetch the largest video recorded at station **ST05** and report its file name and size in bytes.

4. Recommend (2 marks)

Using your own numbers, answer in at most one page:

a) Where should TrailWatch store its photos, clips and recordings, and why?

b) Is MySQL Workbench enough, or is Supabase needed? Give one thing each does better.

c) Which JSON fields would you pull out into normal columns, and why?

Results table (copy into your report and fill in)

Item	Your answer	MySQL BLOB	MySQL path	Supabase
Load all 188 media files	number loaded:	time:	time:	time:
Database size after loading		MB:	MB:	MB:
Retrieve the largest clip	size (bytes):	time:	time:	time:
Q1. TW-00019	station, species, files:	time:		time:
Q2. Gray Wolf, boreal	count:	before / after index:		before / after index:
Q3. low battery readings	count:	before / after index:		before / after index:
Q4. tracks or scat	count:	before / after index:		before / after index:
Q5. largest ST05 video	file and bytes:	time:	time:	time:

Submission Guide:

1. A report of **at most 6 pages** (`Assignment2_YourName_StudentID.pdf`): ER diagram, cleaning notes, the results table, **screenshots** of the photo shown and the clip playing, query plans before and after one index, and your recommendation.
2. Your `.sql` scripts and any loading script, in one ZIP. Marks: design 2, media 3, queries and timing 3, recommendation 2. Late work loses 20% per day for up to three days.
3. You may use AI tools to help you understand the assignment and clarify concepts. However, using AI tools to generate the entire assignment or report will result in a grade of zero. You must demonstrate that you have independently designed, implemented, and executed the assignment on your own computer. You may provide screenshots of your work or submit a screen-recorded video demonstrating the database design, implementation, and successful execution of SQL queries.

Criteria	Excellent	Good	Needs improvement
Design (2)	Sound design (5 tables or fewer), keys correct, choices and cleaning justified	Minor key or cleaning gaps	Missing ERD or keys
Media (3)	All three ways work; photo shown, clip played; times and sizes recorded	One way incomplete	Media not stored or not shown
Queries and timing (3)	Correct answers on both platforms; times before and after an index; plan shown	One or two wrong answers or missing times	Most answers wrong or untimed
Recommendation (2)	Clear choice backed by the student's own numbers	Choice made, little evidence	No clear recommendation